

A PROJECT REPORT

ON

**“Performance Analysis of OFDM Systems
under AWGN, Rayleigh and Nakagami-m
Fading Channels”**

*Submitted in partial fulfilment of the requirements
for the award of the degree of*

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS & COMMUNICATION ENGINEERING



Submitted By:

Sudeep Kumar (22104135027)

Anuradha Kumari (22104135019)

Piyush Kumar (22104135041)

Ujjwal Surya (22104135030)

Under Supervision Of:

Dr. Ravi Ranjan

Asstt. Prof. cum HOD

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

GOVERNMENT ENGINEERING COLLEGE, VAISHALI

(Affiliated to Bihar Engineering University, Patna)

Academic Session: 2025-26

CERTIFICATE OF APPROVAL

The Project work entitled “**Performance Analysis of OFDM Systems under AWGN, Rayleigh and Nakagami-m Fading Channels**” being submitted by:

Name of Student	Registration Number
<i>Sudeep Kumar</i>	22104135027
<i>Piyush Kumar</i>	22104135041
<i>Anuradha Kumari</i>	22104135019
<i>Ujjwal Surya</i>	22104135030

has been examined by us and is hereby approved for the award of the degree of “**Bachelor of Technology (B.Tech)**” in **Electronics and Communication Engineering**, for which it has been submitted.

It is understood that by this approval, the undersigned do not necessarily endorse or approve any statement made, opinion expressed, or conclusion drawn therein, but approve the project work only for the purpose for which it has been submitted.

Internal Examiner

External Examiner

Dr. Ravi Ranjan
Head of Department
Electronics & Communication Engineering
Government Engineering College, Vaishali

Date: _____

Place: Vaishali, Bihar

CERTIFICATE

This is to certify that the project entitled “**Performance Analysis of OFDM Systems under AWGN, Rayleigh and Nakagami-m Fading Channels**” is a bonafide work carried out by:

Name of Student

Registration Number

Sudeep Kumar

22104135027

Piyush Kumar

22104135041

Anuradha Kumari

22104135019

Ujjwal Surya

22104135030

students of **B.Tech in Electronics and Communication Engineering, Government Engineering College Vaishali**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering** during the academic session **2025–26**.

The work embodied in this project report has been carried out under my supervision and guidance. To the best of my knowledge, this work has not been submitted elsewhere for the award of any degree or diploma.

Date: _____

Place: Vaishali, Bihar

Project Supervisor:

Dr. Ravi Ranjan

Asstt. Prof. cum HOD

Internal Examiner

Dr. Tripta

Asstt. Prof.

DECLARATION

We hereby declare that the project report entitled “**Performance Analysis of OFDM Systems under AWGN, Rayleigh and Nakagami-m Fading Channels**” submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering** is an original work carried out by us under the guidance and supervision of **Dr. Ravi Ranjan**, Department of Electronics and Communication Engineering, Government Engineering College Vaishali.

We further declare that the work presented in this report has not been submitted, either in part or in full, to any other University or Institution for the award of any degree, diploma, or certificate. The information and data presented in this report are true to the best of our knowledge and belief.

We understand that any violation of the above declaration may result in the cancellation of the project and appropriate disciplinary action as per the rules and regulations of the institute.

Date: _____

Place: Vaishali, Bihar

Project Team:

Name of Student	Registration Number	Signature
Sudeep Kumar	22104135027	_____
Piyush Kumar	22104135041	_____
Anuradha Kumari	22104135019	_____
Ujjwal Surya	22104135030	_____

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Sudeep Kumar
Anuradha Kumari
Piyush Kumar
Ujjwal Surya

ABSTRACT

Orthogonal Frequency Division Multiplexing (OFDM) is widely used in modern wireless communication systems due to its high spectral efficiency and robustness against multipath fading. It is extensively employed in technologies such as Wi-Fi, LTE, and 5G networks. However, the performance of OFDM systems is affected by channel impairments such as noise, fading, and high Peak-to-Average Power Ratio (PAPR).

This project presents the performance analysis of an OFDM-based wireless communication system using MATLAB and Simulink. The system is implemented using BPSK, QPSK, and 16-QAM modulation techniques and analyzed under AWGN, Rayleigh, and Nakagami-m fading channels. The performance is evaluated using Bit Error Rate (BER) over different Signal-to-Noise Ratio (SNR) values.

A comparative analysis is carried out to study the effect of different modulation schemes and wireless channel conditions on system performance. A Nakagami-m parameter study is also performed to analyse the impact of fading severity. In addition, the PAPR characteristics of the OFDM signal are investigated using Complementary Cumulative Distribution Function (CCDF) curves.

The simulation results show that the AWGN channel provides the best BER performance, while Rayleigh fading produces the highest error rate due to severe multipath effects. Nakagami-m fading offers a more flexible and realistic channel model whose performance depends on the fading parameter m . The study also highlights the trade-off between spectral efficiency and error performance among BPSK, QPSK, and 16-QAM modulation techniques.

Overall, the project provides an effective analysis of OFDM system performance under different wireless channel conditions and modulation schemes.

Keywords: OFDM, BPSK, QPSK, 16-QAM, AWGN, Rayleigh Fading, Nakagami-m Fading, BER, PAPR, MATLAB, Simulink.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
<i>OFDM</i>	<i>Orthogonal Frequency Division Multiplexing</i>
<i>BER</i>	<i>Bit Error Rate</i>
<i>AWGN</i>	<i>Additive White Gaussian Noise</i>
<i>PAPR</i>	<i>Peak-to-Average Power Ratio</i>
<i>FFT</i>	<i>Fast Fourier Transform</i>
<i>IFFT</i>	<i>Inverse Fast Fourier Transform</i>
<i>CP</i>	<i>Cyclic Prefix</i>
<i>SNR</i>	<i>Signal-to-Noise Ratio</i>
<i>QPSK</i>	<i>Quadrature Phase Shift Keying</i>
<i>BPSK</i>	<i>Binary Phase Shift Keying</i>
<i>QAM</i>	<i>Quadrature Amplitude Modulation</i>
<i>CCDF</i>	<i>Complementary Cumulative Distribution Function</i>
<i>PDF</i>	<i>Probability Density Function</i>
<i>LTE</i>	<i>Long Term Evolution</i>
<i>WLAN</i>	<i>Wireless Local Area Network</i>
<i>RF</i>	<i>Radio Frequency</i>
<i>LOS</i>	<i>Line of Sight</i>
<i>NLOS</i>	<i>Non-Line of Sight</i>

1. INTRODUCTION

1.1 Introduction to Wireless Communication

Wireless communication has become an essential part of modern technology and plays a crucial role in enabling communication without physical connections. Over the past few decades, wireless communication systems have evolved significantly to support various applications such as mobile communication, satellite communication, wireless internet, and Internet of Things (IoT) devices. The increasing demand for high data rates, efficient spectrum utilization, and reliable communication has led to the development of advanced wireless communication technologies.

In wireless communication systems, information is transmitted using electromagnetic waves through the air as the transmission medium. However, wireless channels are often affected by various impairments such as noise, interference, and multipath propagation. These impairments can degrade signal quality and increase the probability of transmission errors.

To overcome these challenges, advanced modulation and transmission techniques are used. One of the most important techniques used in modern communication systems is **Orthogonal Frequency Division Multiplexing (OFDM)**. OFDM improves spectral efficiency and reduces the effects of inter-symbol interference caused by multipath propagation.

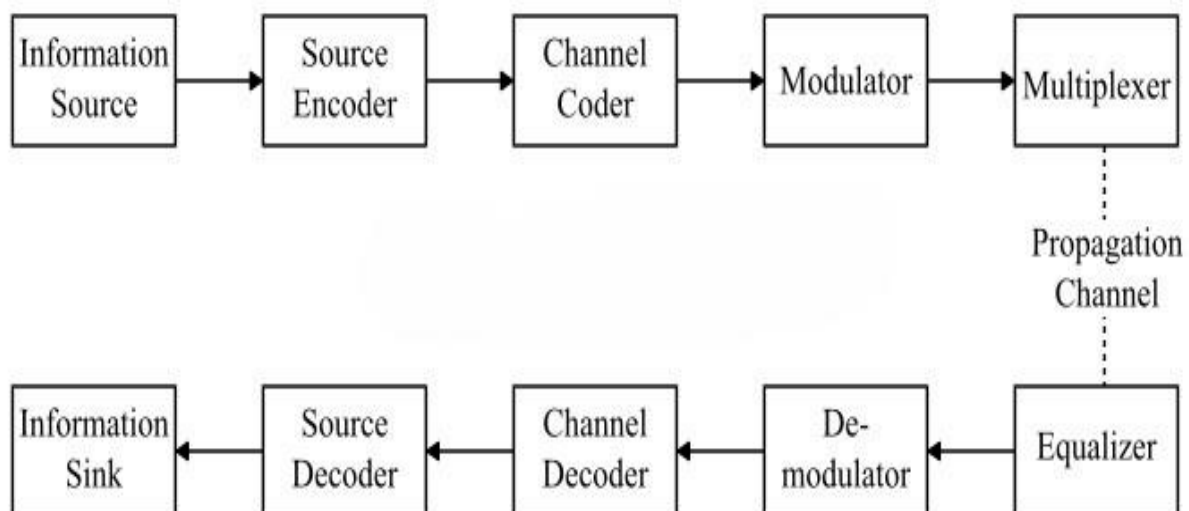


Figure 1 Basic Wireless Communication System

Generation	Technology	Data Rate	Application
1G	Analog Communication	Low	Voice Communication
2G	GSM	Up to 64 kbps	Digital Voice
3G	UMTS	Up to 2 Mbps	Voice & Data
4G	LTE	Up to 100 Mbps	Mobile Broadband
5G	NR	Up to 10 Gbps	Ultra-high-speed communication

Table 1 Evolution of Wireless Communication Technologies

1.2 Overview of OFDM Technology

Orthogonal Frequency Division Multiplexing (OFDM) is a multicarrier modulation technique widely used in modern wireless communication systems. Unlike traditional single-carrier transmission techniques, OFDM divides a high-speed data stream into multiple low-rate parallel data streams and transmits them simultaneously using multiple orthogonal subcarriers.

OFDM also includes the use of a **cyclic prefix**, which helps reduce inter-symbol interference caused by multipath propagation.

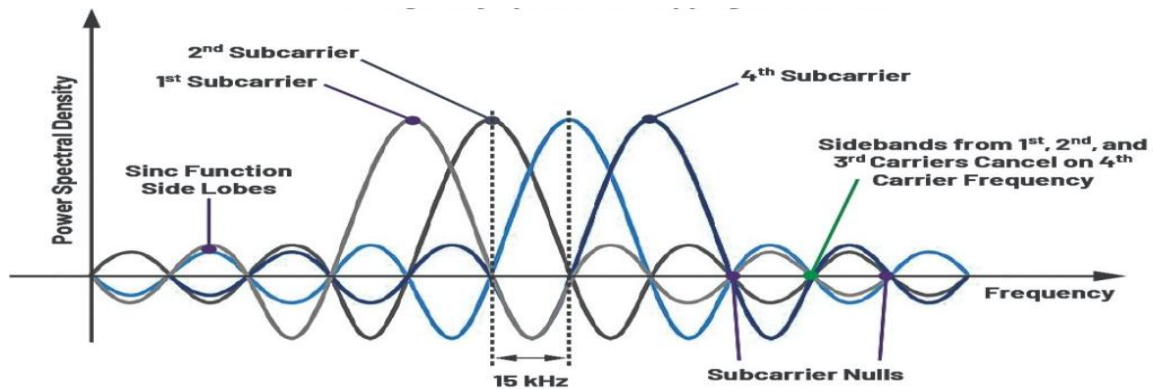


Figure 2 Concept of OFDM Subcarriers

1.3 Problem Statement

Wireless communication systems are significantly affected by noise and fading, which degrade signal quality and increase the probability of transmission errors. Although OFDM effectively reduces the impact of multipath propagation and ISI, its performance is still influenced by channel conditions such as AWGN, Rayleigh fading, and Nakagami-m fading.

Furthermore, OFDM signals exhibit a high Peak-to-Average Power Ratio (PAPR), which reduces power amplifier efficiency and introduces nonlinear distortion.

Therefore, there is a need to analyse the performance of OFDM systems under different channel conditions and modulation schemes while also investigating the PAPR characteristics of the transmitted signal.

1.3 Objectives of the Project

The primary objectives of this project are:

- To design and implement an OFDM-based wireless communication system.
- To implement BPSK, QPSK, and 16-QAM modulation schemes.
- To evaluate OFDM performance under AWGN channels.
- To analyse OFDM performance under Rayleigh fading channels.
- To investigate OFDM performance under Nakagami-m fading channels.
- To compare BER performance for different modulation techniques.
- To study the impact of fading severity using the Nakagami parameter.
- To analyse the PAPR characteristics of OFDM signals.
- To develop and verify the system using MATLAB and Simulink.

1.4 Scope of Work

The scope of this project is limited to the simulation and performance analysis of OFDM-based wireless communication systems. The study focuses on comparing the performance of different modulation techniques under various wireless channel models.

The work includes:

- MATLAB implementation of OFDM transmitter and receiver.
- Simulink modelling of the communication system.
- BER analysis using BPSK, QPSK, and 16-QAM.
- Performance evaluation under AWGN, Rayleigh, and Nakagami-m channels.
- PAPR analysis using CCDF curves.
- Comparative study of modulation schemes and channel conditions.

2. LITERATURE REVIEW

2.1 Review of Existing Work

Several studies have investigated the BER performance of OFDM systems under AWGN and fading channel conditions. Researchers have shown that OFDM significantly improves communication reliability by reducing the effects of Inter-Symbol Interference (ISI) and efficiently utilizing the available spectrum.

Pandey et al. analyzed OFDM systems using BPSK and QPSK modulation schemes under AWGN channels and reported that BPSK achieves lower BER compared to QPSK at low Signal-to-Noise Ratio (SNR) values. Their work demonstrated the importance of modulation selection in determining communication reliability.

Singh and Kumar investigated OFDM performance under Rayleigh fading channels and observed that multipath propagation causes severe signal degradation compared to AWGN channels. Their study highlighted the significance of channel modeling in wireless communication analysis.

Sharma et al. compared BPSK, QPSK, and higher-order modulation schemes such as 16-QAM. The authors concluded that increasing the modulation order improves spectral efficiency and data rate but also increases susceptibility to noise and fading.

Patel et al. studied OFDM performance under Nakagami- m fading channels and demonstrated that the fading parameter m plays a crucial role in determining communication performance. Their results showed that increasing the value of m reduces fading severity and improves BER performance.

Several researchers have also focused on the PAPR problem in OFDM systems. Ali et al. investigated the impact of high PAPR on power amplifier efficiency and concluded that large signal peaks reduce transmitter performance and increase implementation complexity.

Recent studies have employed MATLAB and Simulink for the design and evaluation of OFDM systems. Simulink-based implementations provide a graphical representation of the communication chain and facilitate system-level verification of transmitter and receiver operations.

2.2 Literature Comparison

Table 2 Comparison of Existing Research Works

Author	Modulation Scheme	Channel Model	Performance Metric	Key Observation
Pandey et al.	BPSK, QPSK	AWGN	BER	BPSK achieved lower BER
Singh and Kumar	BPSK	Rayleigh	BER	Fading increased BER significantly
Sharma et al.	BPSK, QPSK, 16-QAM	AWGN	BER	Higher-order modulation improved data rate
Patel et al.	OFDM	Nakagami-m	BER	Higher <i>m</i> improved performance
Ali et al.	OFDM	AWGN	PAPR	High PAPR reduced amplifier efficiency
Choudhary et al.	OFDM	MATLAB/Simulink	BER	Simulink useful for system verification

2.3 Research Gap

From the literature survey, it is observed that most OFDM performance studies primarily focus on BER analysis under AWGN and Rayleigh fading channels. Although some researchers have investigated Nakagami-m fading environments, comprehensive studies involving multiple modulation schemes under different fading conditions remain limited.

Furthermore, many existing works concentrate solely on BER performance without considering the PAPR characteristics of OFDM signals. The combined evaluation of BER performance, Nakagami-m fading analysis, and PAPR characteristics using both MATLAB and Simulink implementations is relatively less explored.

Therefore, there exists a need for a comprehensive OFDM performance study that simultaneously investigates:

- BPSK, QPSK, and 16-QAM modulation schemes

- AWGN, Rayleigh, and Nakagami- m fading channels
- BER performance under different channel conditions
- Effect of the Nakagami fading parameter m
- PAPR characteristics using CCDF analysis
- MATLAB and Simulink-based implementation and verification

2.5 Contribution of Present Work

The present work aims to address the identified research gaps through the following contributions:

1. Development of an OFDM-based wireless communication system using MATLAB and Simulink.
2. Implementation of BPSK, QPSK, and 16-QAM modulation techniques.
3. Performance evaluation under AWGN, Rayleigh, and Nakagami- m fading channels.
4. Comparative BER analysis of different modulation schemes.
5. Investigation of the effect of the Nakagami fading parameter on system performance.
6. Analysis of Peak-to-Average Power Ratio (PAPR) characteristics using CCDF curves.
7. Comparative study of modulation techniques and channel models under practical wireless communication environments.

3. THEORETICAL BACKGROUND

3.1 Orthogonal Frequency Division Multiplexing (OFDM)

Orthogonal Frequency Division Multiplexing (OFDM) is a multicarrier transmission technique in which a high-rate data stream is divided into multiple low-rate parallel streams and transmitted simultaneously over several orthogonal subcarriers. The orthogonality among subcarriers allows them to overlap in the frequency domain without causing interference, thereby improving spectral efficiency.

OFDM is widely used in modern communication systems such as Wi-Fi, LTE, DVB, WiMAX, and 5G due to its robustness against multipath fading and efficient utilization of available bandwidth.

Advantages of OFDM

- High spectral efficiency
- Reduced Inter-Symbol Interference (ISI)
- Robustness against multipath fading
- Efficient implementation using FFT and IFFT
- Suitable for high-speed wireless communication

Applications of OFDM

- Wi-Fi (IEEE 802.11)
- LTE and LTE-Advanced
- Digital Video Broadcasting (DVB)
- WiMAX
- 5G Communication Systems

3.2 Orthogonality in OFDM

The key principle behind OFDM is orthogonality among subcarriers. Two signals are said to be orthogonal if their cross-correlation over a symbol period is zero.

Mathematically,

$$\int_0^T \phi_i(t) \phi_j(t) dt = 0$$

for $i \neq j$
where:

- T = Symbol duration
- $\phi_i(t)$ = i th subcarrier
- $\phi_j(t)$ = j th subcarrier

Due to orthogonality, subcarriers can overlap in the frequency domain without causing Inter-Carrier Interference (ICI).

3.3 FFT and IFFT in OFDM

OFDM systems use Fast Fourier Transform (FFT) and Inverse Fast Fourier Transform (IFFT) for efficient implementation.

IFFT at Transmitter

The IFFT converts frequency-domain symbols into time-domain OFDM signals for transmission.

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi kn/N}$$

where:

- N = Number of subcarriers
- $X(k)$ = Input symbol

FFT at Receiver

FFT converts the received OFDM signal back into the frequency domain for symbol recovery.

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

3.4 Cyclic Prefix (CP)

A Cyclic Prefix is a replica of the end portion of an OFDM symbol appended at the beginning before transmission. The cyclic prefix serves as a guard interval and helps eliminate Inter-Symbol Interference (ISI) caused by multipath propagation.

Advantages of Cyclic Prefix

- Reduces ISI
- Preserves orthogonality
- Simplifies receiver design

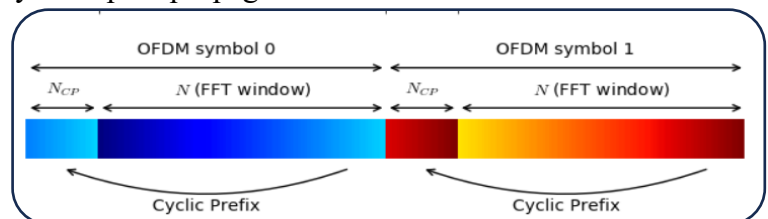


Figure 3: Cyclic Prefix in OFDM

3.5 Binary Phase Shift Keying (BPSK)

Binary Phase Shift Keying (BPSK) is the simplest digital modulation technique in which one bit is represented by one symbol.

The carrier phase changes according to the input bit:

- Bit 0 → Phase 0°
- Bit 1 → Phase 180°

BPSK Signal

$$s(t) = A \cos(2\pi f_c t + \theta)$$

where:

$$\theta = \begin{cases} 0^\circ & \text{for bit 0} \\ 180^\circ & \text{for bit 1} \end{cases}$$

Advantages

- Low BER
- Simple implementation
- Good noise immunity

Disadvantage

- Low spectral efficiency

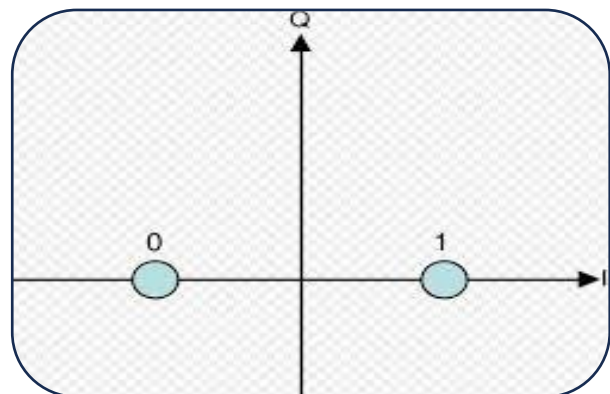


Figure 5 BPSK Constellation Diagram

3.6 Quadrature Phase Shift Keying (QPSK)

Quadrature Phase Shift Keying (QPSK) improves spectral efficiency by transmitting two bits per symbol.

Symbol Mapping

Bits	Phase
00	45°
01	135°
11	225°
10	315°

Advantages

- Doubles data rate compared to BPSK
- Better bandwidth utilization

Disadvantage

- Higher BER than BPSK under identical channel conditions

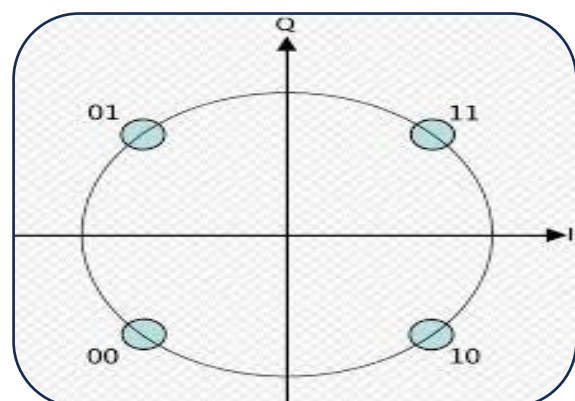


Figure 7 QPSK Constellation Diagram

3.7 16-QAM Modulation

16-Quadrature Amplitude Modulation (16-QAM) combines amplitude and phase modulation to transmit four bits per symbol.

$$\log_2(16) = 4$$

bits are transmitted using a single symbol.

Advantages

- High spectral efficiency
- Increased data rate

Disadvantages

- More susceptible to noise
- Requires higher SNR

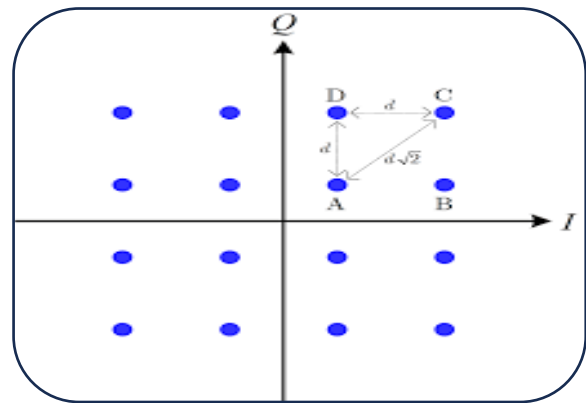


Figure 8 16-QAM Constellation Diagram

Table 3 Comparison of Modulation Schemes

Modulation	Bits per Symbol
BPSK	1
QPSK	2
16-QAM	4

3.8 Additive White Gaussian Noise (AWGN) Channel

The AWGN channel is the simplest channel model used in communication systems. It assumes that the transmitted signal is affected only by additive Gaussian noise.

The received signal is given by:

$$r(t) = s(t) + n(t)$$

where:

- $s(t)$ = Transmitted signal
- $n(t)$ = Gaussian noise

Characteristics:

- Constant power spectral density
- No fading
- Ideal channel model

3.9 Rayleigh Fading Channel

Rayleigh fading models wireless communication environments where there is no direct line-of-sight path between transmitter and receiver.

The received signal experiences rapid amplitude fluctuations due to multipath propagation.

Characteristics

- Severe fading
- Multipath environment
- No LOS component

The probability density function is:

$$f(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}$$

where:

- r = Signal amplitude
- σ^2 = Variance

3.10 Nakagami-m Fading Channel

Nakagami-m fading is a generalized fading model capable of representing a wide range of wireless channel conditions.

Its probability density function is:

$$f(r) = \frac{2m^m}{\Gamma(m)\Omega^m} r^{2m-1} e^{-mr^2/\Omega}$$

where:

- m = Fading parameter
- Ω = Average power
- $\Gamma(m)$ = Gamma function

Table 4 Effect of m

m Value	Fading Condition
0.5	Severe fading
1	Rayleigh fading
2	Moderate fading
4	Mild fading

The Nakagami model provides greater flexibility than Rayleigh fading and can represent various practical wireless environments.

3.11 Bit Error Rate (BER)

Bit Error Rate (BER) is one of the most important performance metrics used to evaluate digital communication systems.

BER is defined as:

$$BER = \frac{\text{Number of Error Bits}}{\text{Total Number of Transmitted Bits}}$$

A lower BER indicates better communication system performance.

Factors Affecting BER

- Modulation technique
- Channel conditions
- Noise level
- Signal-to-Noise Ratio (SNR)

3.12 Peak-to-Average Power Ratio (PAPR)

One of the major drawbacks of OFDM systems is the high Peak-to-Average Power Ratio (PAPR). PAPR is defined as:

$$PAPR = \frac{\max |x(n)|^2}{E[|x(n)|^2]}$$

where:

- $\max |x(n)|^2$ = Peak signal power
- $E[|x(n)|^2]$ = Average signal power

Effects of High PAPR

- Reduced power amplifier efficiency
- Nonlinear distortion
- Increased implementation complexity

PAPR Analysis

The PAPR characteristics of OFDM signals are commonly evaluated using the Complementary Cumulative Distribution Function (CCDF).

CCDF represents the probability that the PAPR exceeds a specified threshold.

4. SYSTEM DESIGN AND IMPLEMENTATION

The proposed communication system consists of three major sections:

- OFDM Transmitter
- Wireless Channel
- OFDM Receiver

The transmitter converts binary information into OFDM symbols and transmits them through the wireless channel. The receiver performs the inverse operations to recover the transmitted information and evaluate system performance.

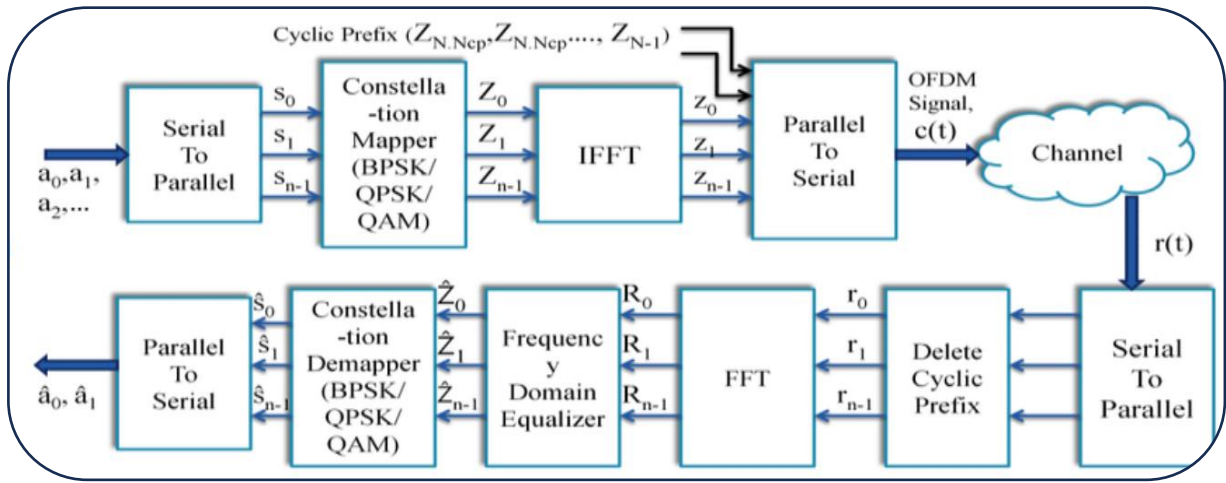


Figure 9 Proposed OFDM System Architecture

4.1 OFDM Transmitter Design

The transmitter is responsible for generating and preparing the data for transmission.

The transmitter performs the following operations:

Step 1: Random Bit Generation: A random binary sequence is generated using MATLAB.

Step 2: Digital Modulation: The generated bits are mapped using:

- BPSK
- QPSK
- 16-QAM

Step 3: Serial-to-Parallel Conversion: Modulated symbols are grouped into OFDM blocks.

Step 4: IFFT Operation: IFFT converts frequency-domain symbols into time-domain OFDM.

Step 5: Cyclic Prefix Insertion: A cyclic prefix is added to reduce the effects of multipath propagation and ISI.

4.2 OFDM Receiver Design

The receiver performs inverse operations of transmitter to recover the transmitted information.

Step 1: Cyclic Prefix Removal: Cyclic prefix is removed from the received OFDM symbol.

Step 2: FFT Operation: FFT converts the received time-domain signal back into the frequency domain.

Step 3: Parallel-to-Serial Conversion: Subcarrier data are converted into a serial data stream.

Step 4: Demodulation: The received symbols are demodulated using the corresponding modulation scheme.

Step 5: BER Calculation: Received bits are compared with transmitted bits to calculate BER.

4.3 MATLAB Implementation

The complete OFDM communication system is implemented using MATLAB.

The implementation flow is:

- Generate random binary data.
- Apply BPSK/QPSK/16-QAM modulation.
- Convert serial data to parallel form.
- Perform IFFT operation.
- Add cyclic prefix.
- Pass signal through selected channel.
- Remove cyclic prefix.
- Apply FFT operation.
- Perform demodulation.
- Calculate BER.
- Repeat for different SNR values.
- Generate BER curves.

4.4 Simulink Implementation

To provide a graphical representation of the communication system, the OFDM model is also implemented using Simulink.

The Simulink model allows visualization of transmitter, channel, and receiver operations.

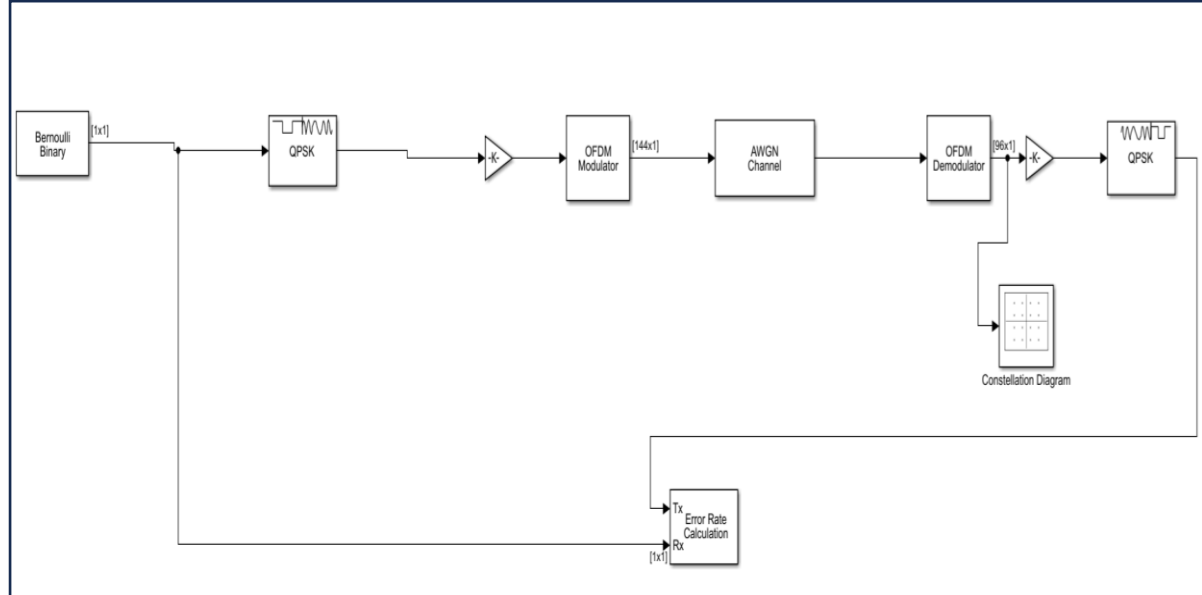


Figure 10 Complete Simulink OFDM Model

4.5 PAPR Measurement Method

The Peak-to-Average Power Ratio (PAPR) is evaluated using the generated OFDM signal.

PAPR is calculated as:

$$PAPR = \frac{\max |x(n)|^2}{E[|x(n)|^2]}$$

where:

- Peak Power = $\max |x(n)|^2$
- Average Power = $E[|x(n)|^2]$

The Complementary Cumulative Distribution Function (CCDF) is used to analyze the probability that PAPR exceeds a specified threshold.

5. RESULTS AND PERFORMANCE ANALYSIS

5.1 BER Performance under AWGN Channel

The AWGN channel represents an ideal communication environment where the transmitted signal is affected only by additive Gaussian noise.

BER performance is evaluated for BPSK, QPSK, and 16-QAM over SNR range of 0–20 dB.

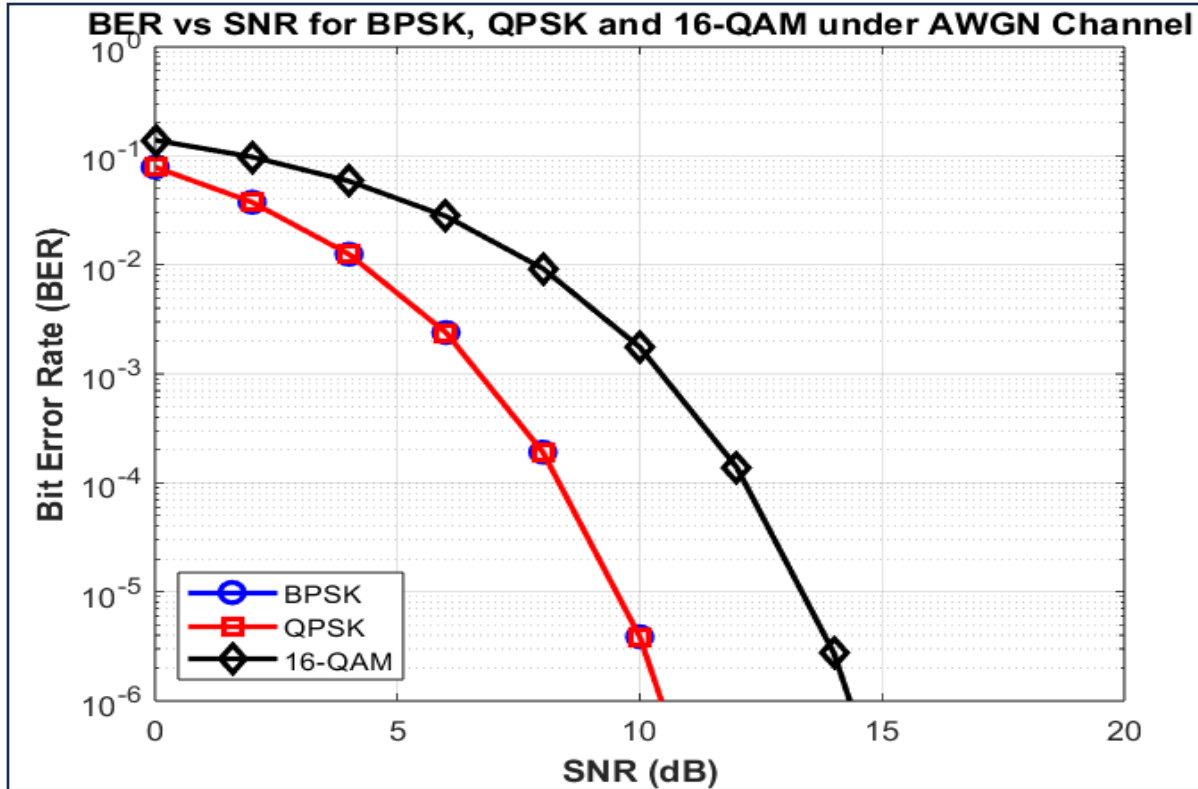


Figure 11 BER vs SNR for BPSK, QPSK and 16-QAM under AWGN Channel

Observations

- BER decreases as SNR increases.
- BPSK exhibits the lowest BER.
- QPSK provides moderate BER performance.
- 16-QAM requires higher SNR to achieve acceptable BER.
- AWGN provides the best overall communication performance.

5.2 BER Performance under Rayleigh Fading Channel

The Rayleigh channel models multipath propagation without a direct line-of-sight component.

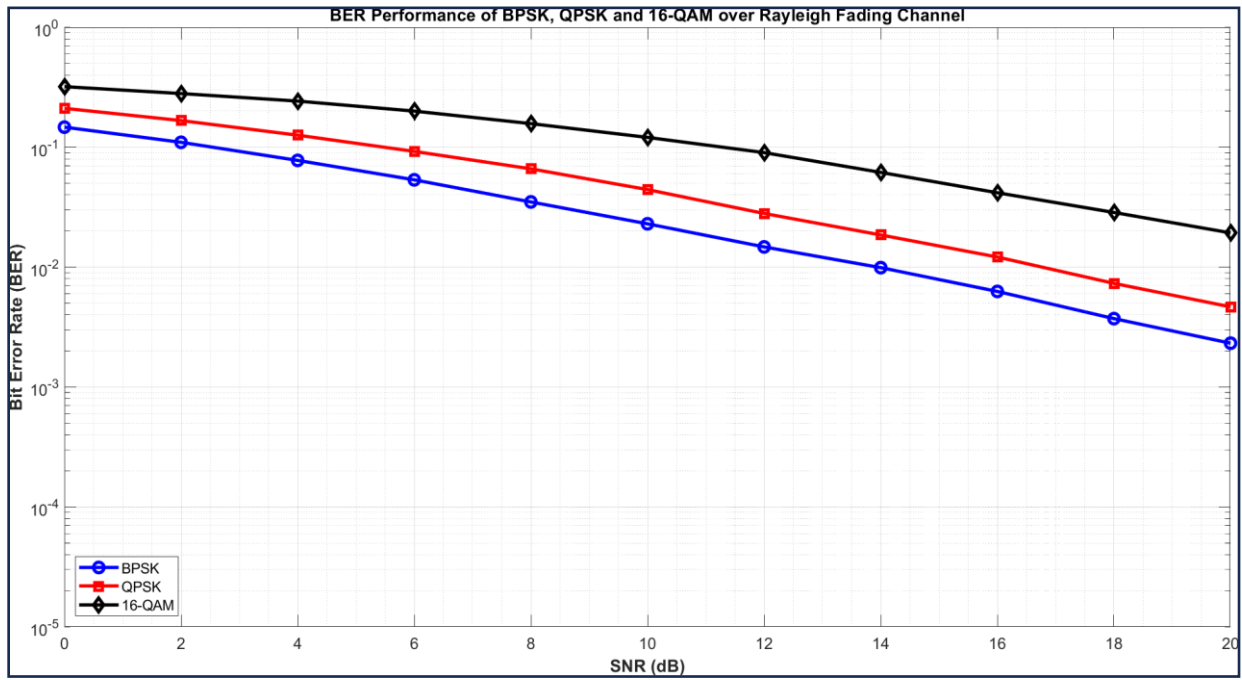


Figure 12 BER vs SNR for BPSK, QPSK and 16-QAM under Rayleigh Channel

Observations

- BER performance degrades significantly compared to AWGN.
- Multipath fading introduces signal fluctuations.
- BPSK remains the most reliable modulation technique.
- 16-QAM is highly sensitive to fading effects.

5.3 BER Performance under Nakagami-m Fading Channel

Nakagami-m fading provides a generalized model capable of representing different wireless propagation environments.

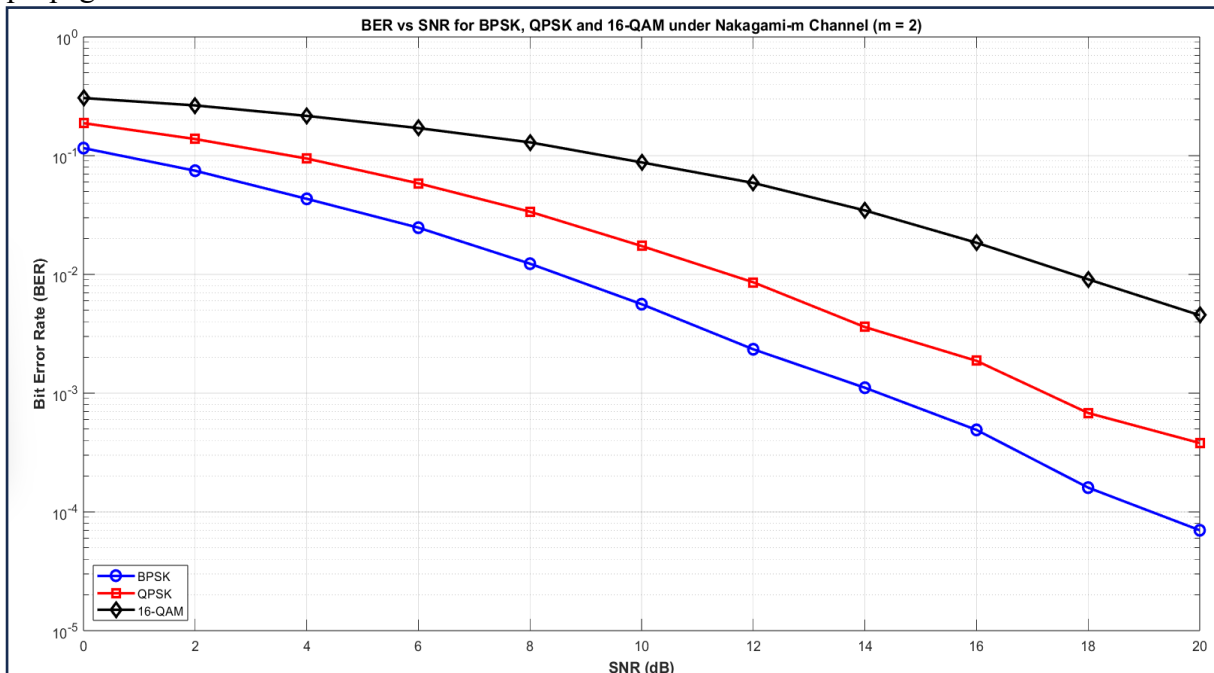


Figure 13 BER vs SNR for BPSK, QPSK and 16-QAM under Nakagami-m Channel

Observations

- Performance depends on fading parameter m .
- BER decreases as m increases.
- Nakagami fading provides more flexible channel modeling.
- Results lie between AWGN and Rayleigh channels.

5.4 Comparison of Modulation Techniques

A comparative study is performed to evaluate the performance of BPSK, QPSK, and 16-QAM under identical channel conditions.

Table 5 Comparison of Modulation Techniques

Modulation	Bits/Symbol	BER Performance
BPSK	1	Best
QPSK	2	Moderate
16-QAM	4	Lowest Reliability

Observation

As modulation order increases:

- Spectral efficiency increases.
- Data rate increases.
- BER performance decreases.

This demonstrates the trade-off between reliability and bandwidth efficiency.

5.5 Comparison of Wireless Channel Models

A comparison is performed among AWGN, Rayleigh, and Nakagami- m channels.

Table 6 Comparison of Wireless Channel Models

Channel	Performance
AWGN	Best
Nakagami- m	Moderate
Rayleigh	Worst

Observation

AWGN provides ideal performance due to the absence of fading. Rayleigh fading causes severe signal degradation, while Nakagami- m provides adjustable fading conditions through parameter m .

5.6 Effect of Nakagami Parameter (m)

The influence of fading severity is analyzed using different values of the Nakagami parameter.

Values considered:

- $m = 0.5$
- $m = 1$
- $m = 2$
- $m = 4$

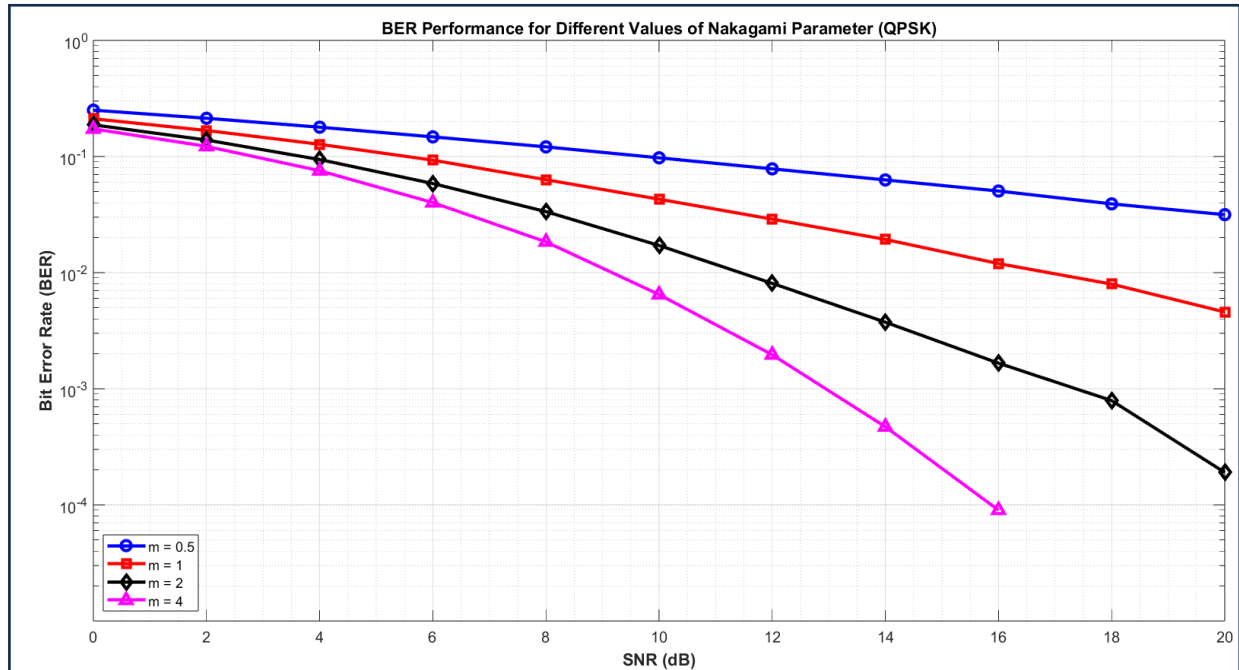


Figure 15 BER Performance for Different Values of m QPSK

Observations

- $m = 0.5$ results in severe fading.
- $m = 1$ corresponds to Rayleigh fading.
- Increasing m improves BER performance.
- $m = 4$ provides the best performance among the selected values.

Table 7 Effect of Nakagami Parameter

m Value	Channel Condition
0.5	Severe Fading
1	Rayleigh Equivalent
2	Moderate Fading
4	Mild Fading

5.7 PAPR Analysis

One of the major limitations of OFDM systems is high Peak-to-Average Power Ratio (PAPR).

PAPR analysis is performed using Complementary Cumulative Distribution Function (CCDF).

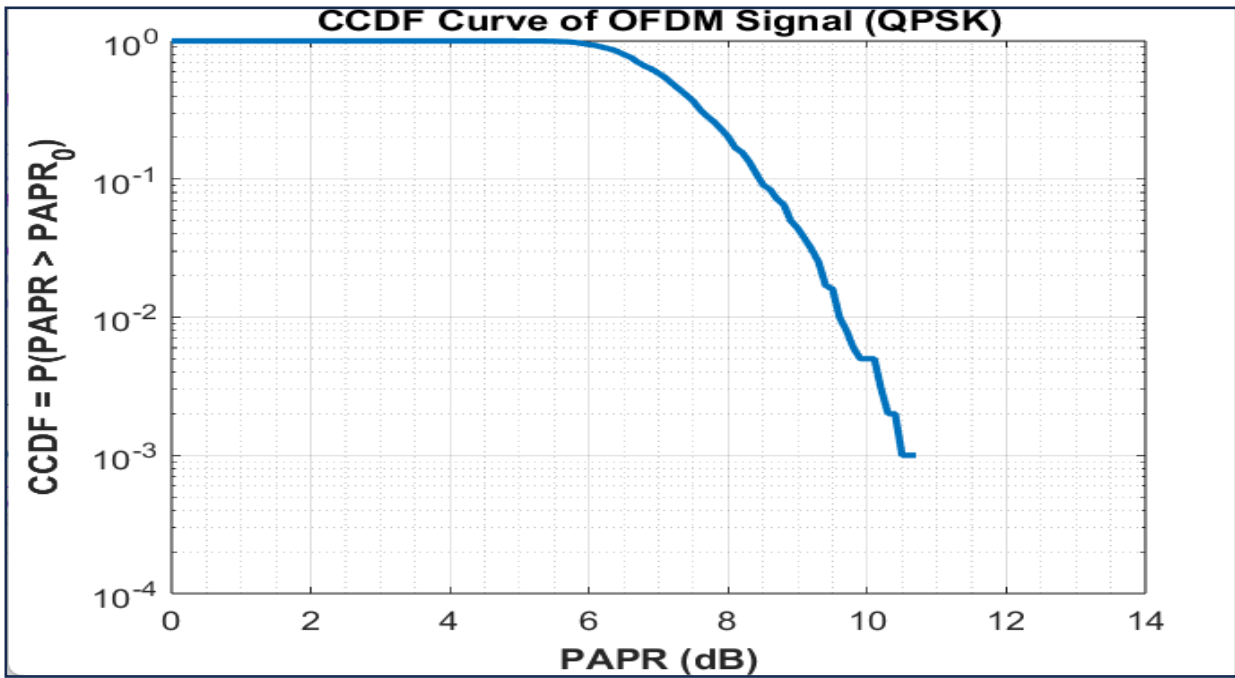


Figure 17 CCDF Curve of OFDM Signal

Observations

- OFDM signals exhibit high PAPR values.
- Large peaks occur due to the addition of multiple subcarriers.
- High PAPR reduces power amplifier efficiency.
- PAPR increases with higher-order modulation schemes.

5.8 Comparative PAPR Analysis

The PAPR characteristics of different modulation techniques are compared.

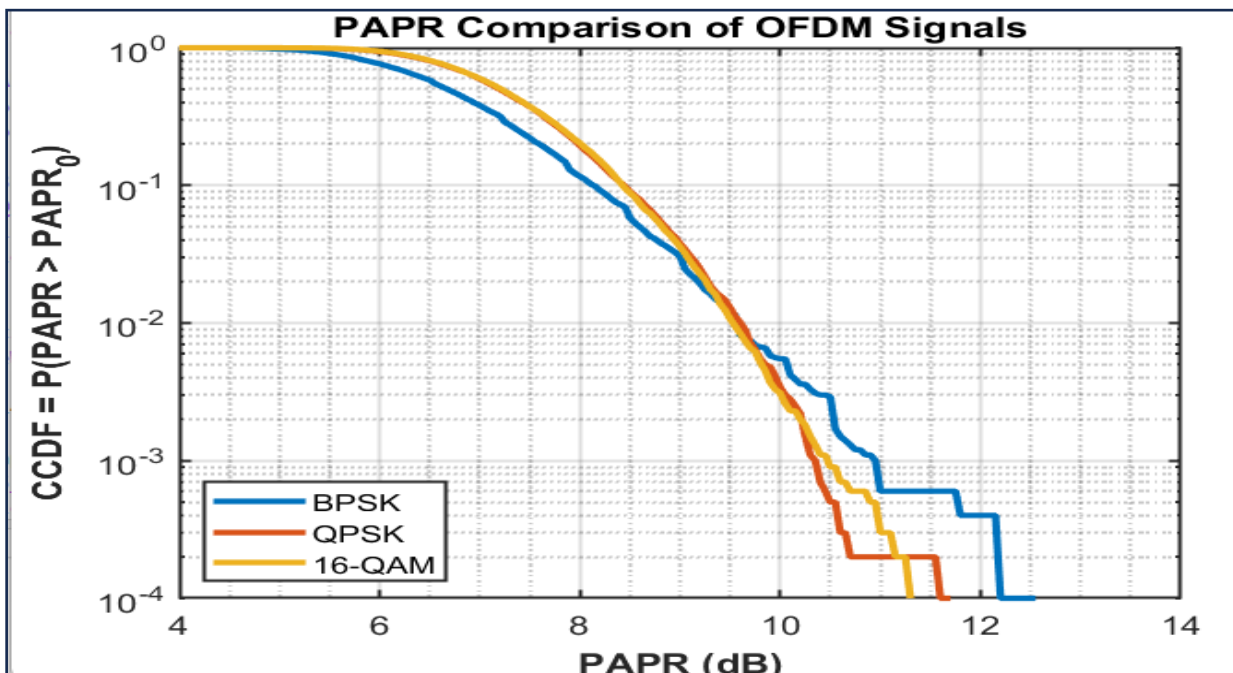


Figure 18 PAPR Comparison for BPSK, QPSK and 16-QAM

Table 8 Comparative PAPR Analysis

Modulation	Relative PAPR
BPSK	Lowest
QPSK	Moderate
16-QAM	Highest

Observation

As modulation complexity increases, PAPR tends to increase, making transmitter design more challenging.

Table 9 Overall Performance Comparison

Parameter	BPSK	QPSK	16-QAM
Bits/Symbol	1	2	4
Spectral Efficiency	Low	Medium	High
BER Performance	Best	Moderate	Lowest
Data Rate	Low	Medium	High
PAPR	Low	Moderate	High

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project presented the design and performance analysis of an Orthogonal Frequency Division Multiplexing (OFDM)-based wireless communication system under different wireless channel conditions. The complete system was implemented using MATLAB and Simulink to evaluate the behaviour of OFDM in practical communication environments.

Three digital modulation techniques, namely BPSK, QPSK, and 16-QAM, were implemented and analyzed to study the trade-off between spectral efficiency and error performance. The system was evaluated under AWGN, Rayleigh, and Nakagami- m fading channels using Bit Error Rate (BER) as the primary performance metric. In addition, the Peak-to-Average Power Ratio (PAPR) characteristics of the OFDM signal were investigated using Complementary Cumulative Distribution Function (CCDF) analysis.

The simulation results demonstrated that the BER performance improves with increasing Signal-to-Noise Ratio (SNR) for all modulation schemes and channel models. Among the considered modulation techniques, BPSK achieved the lowest BER and provided the highest reliability, while 16-QAM offered higher data rates at the cost of increased error probability. QPSK provided a balanced compromise between reliability and spectral efficiency.

A comparative analysis of channel models revealed that the AWGN channel provides the best communication performance due to the absence of fading effects. The Rayleigh fading channel exhibited the highest BER because of severe multipath propagation. The Nakagami- m fading channel provided a more flexible and realistic fading model, and the study showed that increasing the fading parameter m improves system performance by reducing fading severity.

The PAPR analysis highlighted one of the major limitations of OFDM systems. The results indicated that OFDM signals exhibit high peak power values due to the simultaneous transmission of multiple subcarriers, which can reduce power amplifier efficiency and introduce nonlinear distortion. The CCDF curves provided valuable insights into the power characteristics of OFDM signals under different modulation schemes.

Overall, the objectives of the project were successfully achieved. The developed MATLAB and Simulink models effectively demonstrated the impact of modulation techniques and wireless channel conditions on OFDM system performance.

6.2 Future Scope

Although the proposed OFDM system demonstrated satisfactory performance under different modulation schemes and channel conditions, several enhancements can be incorporated in future work to further improve system performance and practical applicability.

One possible extension of this work is the implementation of higher-order modulation techniques such as 64-QAM and 256-QAM to achieve greater spectral efficiency and higher data transmission rates. The effect of these modulation schemes on BER and PAPR performance can also be investigated.

The performance of OFDM systems can be significantly improved by incorporating channel coding techniques such as Convolutional Codes, Turbo Codes, and Low-Density Parity Check (LDPC) Codes. These coding methods can enhance error correction capability and improve communication reliability in noisy and fading environments.

Another important area for future research is the integration of Multiple Input Multiple Output (MIMO) technology with OFDM. MIMO-OFDM systems offer higher capacity, improved reliability, and increased spectral efficiency, making them suitable for advanced wireless communication networks.

Since high PAPR remains one of the major drawbacks of OFDM systems, future work can focus on implementing PAPR reduction techniques such as Selective Mapping (SLM), Partial Transmit Sequence (PTS), and Clipping and Filtering methods. These techniques can improve power amplifier efficiency and overall transmitter performance.

Further studies may also incorporate practical channel estimation and equalization techniques to improve receiver performance under realistic wireless channel conditions. Additionally, the developed OFDM framework can be extended for performance evaluation in emerging communication technologies such as 5G New Radio (NR), Massive MIMO systems, Internet of Things (IoT) networks, and future 6G wireless communication systems.

In conclusion, the proposed work serves as a strong foundation for advanced research in wireless communication systems and can be extended to address several practical challenges associated with modern high-speed communication networks.

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